Scale, proportions, circles, and review

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 1248 to the nearest hundred.

2. Round 12.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

6. Find 140% of 30 points.

7. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

8. Find 9.6 ÷ 8.

9. Solve x + 14 = 31.

10. A rectangle is 7 cm by 3 cm. Find its area.

11. Find the mean of 1, 4, 5, 6.

12. Find the median of 10, 3, 5, 7, 2, 9.

13. Find the range of 0, 2, 5, 10.

14. Find the MAD of 2, 5, 5, 8.

15. A map uses 1 cm for 50 km. A route is 2 cm long. Find the actual distance.

16. Maya finds area from a diameter of 14 cm by writing $\pi \times 14^2 = 196\pi$. Explain and repair it.

17. Compute (−3) × 4 and (−12) ÷ (−3).

18. A price rises from $20 to $25. Find the percent increase.

CHALLENGE / Explain your plan and check the result

C1. An elevator starts on floor −3, travels up 8 floors, then down 6. Find its final floor and total floors traveled. Compare the final position with its distance from zero; explain why neither equals total travel.

C2. A drawing shows a triangle with base 8 cm and height 5 cm. A copy has base 12 cm and is scaled uniformly. Find the copy’s height and area. A student multiplies 20 square cm by 1.5 and gets 30. Explain the missing step.